

Yu Xu

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EDUCATION

University of Manchester

Manchester, UK

MSc Robotics

Sep 2025 – Dec 2026

- Academic and project work in robotics software, perception, manipulation and autonomous systems.
- Developed ROS2-based components and validated sensing-to-control workflows through simulation and lab testing.

The University of Hong Kong

Hong Kong

Exchange, Computer Science

Sep 2024 – Dec 2024

- Applied CNNs, data preprocessing and hyperparameter tuning to noisy image datasets.
- Built reproducible training and evaluation workflows in Python and TensorFlow.

Xiamen University

Xiamen, China

BEng Digital Media Technology

Sep 2021 – Jun 2025

- Developed a broad foundation in computer science and digital media engineering through project-based software and intelligent-systems development.

EXPERIENCE

DesignLibro

Shenzhen, China

AI Engineering Intern

Dec 2024 – May 2025

- Analysed user logs to compare HLS and TUTK streaming solutions, and presented findings to support technical evaluation of the in-house approach.
- Built a Streamlit + SQL dashboard with dynamic time filters and user deduplication logic, enabling non-technical stakeholders to self-serve device-usage insights without recurring analyst support.
- Established labelling guidelines and data-quality checks for multimodal pet-camera datasets, improving consistency of training and evaluation data for behaviour-recognition tasks.
- Reviewed edge cases, refined labels and organised evaluation sets to support iteration of behaviour-recognition workflows.
- Conducted compatibility checks and Locust load testing, identifying bottlenecks for platform optimisation.

PROJECTS

RAG-based Historical Dialogue System with Personality Modelling

Project

Dec 2024 – May 2025

- Built a retrieval-augmented dialogue system by collecting and cleaning historical text, generating BERT/RoBERTa embeddings and storing them in Weaviate.
- Developed backend orchestration for retrieval and response generation, improving factual consistency in character-based dialogue.
- Integrated GPT-SoVITS with asynchronous inference workflows to reduce latency and improve interaction quality.

Robotic Systems Design Project — Vision Software Module

Project (Ongoing)

2025 – Present

- Developed the vision software module of a ROS2-based mobile manipulation system for detection and localisation in pick-and-place tasks.
- Built ROS2 nodes for image processing, detection publishing and debugging to support integration with perception, planning and control modules.
- Implemented structured perception outputs and developer tooling to improve testing, reproducibility and cross-team integration.

PetCare AI

Project

Jun 2023 – Jul 2024

- Developed a Django backend for an AI-assisted pet-care consultation system using BERT, GPT-based semantic understanding and multi-turn retrieval.
- Improved intent handling and response relevance compared with a rule-based system.

TECHNICAL SKILLS

Programming: Python, C++, SQL, MATLAB, JavaScript

ML / AI: PyTorch, TensorFlow, scikit-learn, Transformers, OpenCV, Weaviate

Frameworks / Tools: ROS2, Flask, Django, Streamlit, Git, Locust

Other: Data pipelines, RAG systems, vector search, multimodal workflows, backend prototyping, benchmarking, technical reporting